



Master Presenter's Manual

for

Developing ASP.NET MVC Web Applications

Curriculum Code: OV-1509

***Effective from: April, 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	April 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Understand the basics of Web applications with ASP.NET
- Build a Web Forms application
- Understand ASP.NET Core
- Understand ASP.NET MVC and ASP.NET Core MVC
- Build a Web application from an empty template
- Implement an MVC Web application
- Implement a Core MVC Web application
- Implement a data-layer project referenced from the MVC and Razor Page applications
- Implement data handling in ASP.NET MVC with a code-first database using Entity Framework Core
- Add and modify models, views and controllers to perform CRUD operations against the database
- Use client-side and server-side validation
- Secure the Controllers, Actions, and View content with authorization and roles
- Understand how to create Single Page Applications

2. Information on Module Allocation

Module	Online (No. of Hours)	Skill Code
ASP.NET	40	1340

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 ASP.NET-TL1	Session 2 ASP.NET-TL2	Session 3 ASP.NET-TL3	Session 4 ASP.NET-TL4
2	Session 5 ASP.NET-TL5	Session 6 ASP.NET-TL6	Session 7 ASP.NET-TL7	Session 8 ASP.NET-TL8
3	Session 9 ASP.NET-TL9	Session 10 ASP.NET-TL10	Session 11 ASP.NET-TL11	Session 12 ASP.NET-TL12
4	Session 13 ASP.NET-TL13	Session 14 ASP.NET-TL14	Session 15 ASP.NET-TL15	Session 16 ASP.NET-TL16
5	Session 17 ASP.NET-TL17	Session 18 ASP.NET-TL18	Session 19 ASP.NET-TL19	Session 20 ASP.NET-TL20

ASP.NET: Developing ASP.NET Core MVC Applications
TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	ASP.NET – TL1	All the topics as listed below from Session 1 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 1 – Introduction to ASP.NET and ASP.NET Core</u> • Explain ASP.NET Framework and its history • Explain ASP.NET Page Lifecycle and Lifecycle Events • Explain ASP.NET Features and its Uses • Explain ASP.NET Core • List ASP.NET Core Advantages • Describe ASP.NET Core Exceptions and Static Files • Identify how to choose between ASP.NET and ASP.NET Core • Describe the features of ASP.NET Core 7.0	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 1 XP - Session 1 TG - Session 1
2	ASP.NET – TL2	All the topics as listed below from Session 2 and 3 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 2 - Working with ASP.NET Web Forms, Controls, and Events</u> • Explain Web application development and Web Forms • Identify types of Web Forms • Describe event handling in ASP.NET • List and describe various types of controls in ASP.NET • Explain Web API and Web security	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 2 & 3 XP - Session 2 & 3 TG - Session 2 & 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 3 – Working with ADO.NET and Entity Framework</u> - Describe ADO.NET - Explain Entity Framework - Describe data handling in ASP.NET MVC with a code-first database	
3	ASP.NET – TL3	All the topics as listed below from Session 4 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 4 – Client-Side Development Using ASP.NET Core MVC</u> - Describe layouts in ASP.NET Core MVC - Explain implementing styles in ASP.NET Core MVC applications - Explain Data Annotations - Describe routing - Explain dependency injection - Identify the process to create a Single Page Application - Describe client-side validation and server side validation	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 4 XP - Session 4 TG - Session 4
4	ASP.NET – TL4	The Try It Yourself questions of Session 1 to Session 4 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 1-4
5	ASP.NET – TL5	All the topics as listed below from Session 5 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. Session 5 – More on ASP.NET MVC and Core MVC - Explain Role-based and View-based authorization - Describe ASP.NET Selectors - Explain ASP.NET Helpers - List Action Filters - Identify usage of Apply Action	SG - Session 5 XP - Session 5 TG - Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		Filter and Custom Filter - Explain ASP.NET MVC Security - Describe Views and Partial Views in MVC	
6	ASP.NET – TL6	All the topics as listed below from Session 6 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 6 – Action Methods and Advanced Concepts in MVC</u> - Explain Action Methods in MVC - Compare Web API and SOAP - Describe Route Config Declaration - Describe the process for sharing data for ASP.NET Core MVC - Explain Bundling and Minification in MVC - Explain Areas in MVC	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 6 XP - Session 6 TG - Session 6
7	ASP.NET – TL7	All the topics as listed below from Session 7 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 7 – Enhancements in ASP.NET Core</u> - Describe .NET Core MVC and Razor page Enhancements. - Explain MVC Model Binding Enhancements - Describe Text.Json - Describe dump debugging - Outline Performance improvements in recent versions - Identify the use of Garbage Collection	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 7 XP - Session 7 TG - Session 7
8	ASP.NET – TL8	The Try It Yourself questions of Session 5 to Session 7 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 5-7
9	ASP.NET – TL9	All the topics as listed below from Session 8 of <i>Web Programming</i>	<u>Web Programming</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
		<i>Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 8 – .NET Core Architecture and Kestrel Web Server implementation</u> - Describe .NET Core architecture - Explain the execution of Main method - Describe the Kestrel server - Describe the OWIN architecture - Explain REST API architecture - Explain the difference between GraphQL API and REST API	<u>Using ASP.NET CORE and MVC</u> SG - Session 8 - XP - Session 8 - TG - Session 8
10	ASP.NET – TL10	All the topics as listed below from Session 9 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 9 – Onion Architecture in ASP.NET Core - I</u> - Explain inversion of control, dependency inversion principle, and Onion Architecture - Describe the Onion Architecture layers - Define Dependency Inversion Principle (DIP) - Explain the implementation of data access and repositories in the infrastructure layer - Explain advantages of the Onion Architecture	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 9 - XP - Session 9 - TG - Session 9
11	ASP.NET – TL11	All the topics as listed below from Session 10 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 10 – Onion Architecture in ASP.NET Core - II</u> - Explain project structure for Onion Architecture - Describe the process of implementation of the four	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 10 - XP - Session 10 - TG - Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
		projects – Domain layer, Infrastructure layer, Service layer, and Onion Architecture Web API - List the security considerations - Explain setting up monitoring and tracing for error detection and performance analysis	
12	ASP.NET – TL12	The Try It Yourself questions of Session 8 to Session 10 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 8-10
13	ASP.NET – TL13	All the topics as listed below from Session 11 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 11 – Overview of Fluent Model and Implementing AutoMapper in ASP.NET Core MVC</u> - Explain Fluent API and Fluent model - Describe benefits of Fluent API and Fluent Model - Outline the procedure to perform operations such as Create, Edit, Delete, and Update (CRUD) using Fluent API with ASP.NET Core MVC - Define AutoMapper - List reasons for using AutoMapper - Explain the installation and configuration process for AutoMapper - Describe use of AutoMapper in an application	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 11 XP - Session 11 TG - Session 11
14	ASP.NET – TL14	All the topics as listed below from Session 12 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 12 – .NET Core Token Authentication in Web Applications</u> Describe token-based	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 12 XP - Session 12 TG - Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		authentication • Explain the process to validate tokens in ASP .NET core • Explain JSON Web Token • Identify how to set up Token Authentication in .NET Core • Elaborate on Token Generation and Issuing • Explain how to use tokens obtained from identity providers in applications. • Describe Token-Based Authentication in MVC/Web Applications	
15	ASP.NET – TL15	The Try It Yourself questions of Session 11 and Session 12 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 11 & 12
16	ASP.NET – TL16	All the topics as listed below from Session 13 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 13 – Deployment and Unit of Work Patterns in ASP.NET Core</u> • Describe repository pattern • List advantages of repository pattern • Explain the difference between repository pattern and repository pattern with unit of work • List benefits of unit of work	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 13 XP - Session 13 TG - Session 13
17	ASP.NET – TL17	All the topics as listed below from Session 14 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 14 – User Login and ASP.NET Core Identity</u> • Describe User Identity and Login • Explain Password Security • Describe Session State in	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 14 XP - Session 14 TG - Session 14

Session No.	Session Title	Session Details	Deliverables' Mapping
		ASP.NET Core Elaborate on User Identity and Security with Example Application	
18	ASP.NET – TL18	The Try It Yourself questions of Session 13 and Session 14 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 13 & 14
19	ASP.NET – TL19	All the topics as listed below from Session 15 of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session. <u>Session 15 – Publishing and Deploying ASP.NET Core Applications</u> - Describe Web Servers - Explain the Deployment Process in Visual Studio 2022 - Describe IIS configuration - Explain the process to deploy ASP .NET Core and ASP.NET MVC - Explain the installation procedure for Windows features and .NET core Hosting Bundle	<u>Web Programming Using ASP.NET CORE and MVC</u> SG - Session 15 XP - Session 15 TG - Session 15
20	ASP.NET – TL20	The Try It Yourself questions of Session 15 and Appendix of <i>Web Programming Using ASP.NET CORE and MVC</i> book should be covered in this session.	<u>Web Programming Using ASP.NET CORE and MVC</u> Session 15

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ **Deliverables for Student:**

Module	Deliverable
Developing ASP.NET Core MVC Applications	Kit Code: OV-1509-KIT01-01-A Kit Name: Kit of 1 title for OV-1509 Dev ASPNET MVC Web Book Code: AINNASPCR10224E000 Book Name: Web Programming using ASPNET CORE and MVC

Deliverables for Faculty:

Module	Deliverable
Developing ASP.NET Core MVC Applications	Kit Code: OV-1509-KIT01-01-A Kit Name: Kit of 1 title for OV-1509 Dev ASPNET MVC Web Book Code: AINNASPCR10224E000 Book Name: Web Programming using ASPNET CORE and MVC
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Pro ASP.NET Core 7, Tenth Edition by Adam Freeman
➤ Architecting ASP.NET Core Applications - Third Edition by By Carl-Hugo Marcotte
➤ Building Web Applications with C# and .NET: A Complete Reference By Dudley Gill

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for PASS

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for CREDIT and

Overall Weighted Marks $\geq 75\%$ qualifies for DISTINCTION

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50

Duration: 60 minutes

Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Developing ASP.NET Core MVC Applications	50

Final Examination (R1)

Marks: 50

Duration: 120 minutes

Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Developing ASP.NET Core MVC Applications	50

Evaluation Details

Practical – 50%

OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10
Microsoft Visual Studio 2022 Professional Edition

8.2 Recommended Hardware

Intel Core i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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